GNR 638: Machine Learning for Remote Sensing – II

Assignment 2

Transfer Learning on AID Dataset

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1 Introduction

This report documents the design, implementation, and evaluation of various transfer learning paradigms using pre-trained Convolutional Neural Networks (CNNs) on the Aerial Images Dataset (AID), as required by the assignment for GNR 638.

The primary objective of this project was to analyze how different Transfer Learning models adapt to specialized remote sensing imagery. To achieve a comprehensive understanding, we evaluated three distinct architectural paradigms:

- **ResNet50**: A standard deep residual network.
- **EfficientNet-B0**: A lightweight, compound-scaled mobile architecture.
- **ConvNeXt-Tiny**: A modernized pure-convolutional architecture inspired by Vision Transformers.

The report covers five rigorous experimental scenarios:

1. **Scenario 1:** Linear Probe Transfer (Frozen Backbone)
2. **Scenario 2:** Fine-Tuning Strategies (Layer-wise unfreezing comparisons)
3. **Scenario 3:** Few-Shot Learning Analysis (Data scarcity resilience)
4. **Scenario 4:** Corruption Robustness Evaluation (OOD performance)
5. **Scenario 5:** Layer-Wise Feature Probing (Analyzing hierarchical feature emergence)

All models were evaluated using appropriate metrics, including Validation Accuracy, Overfitting Gaps, Relative Drops, and visual inspections through Confusion Matrices and Dimensionality Reduction (PCA) visualizations.

2 Hardware Limitations and Design Challenges

During the execution of these scenarios, specific hardware constraints influenced our approach, adding valuable engineering context to the experiments.

Memory Bottleneck with ConvNeXt-Tiny: It was observed that while ResNet50 and EfficientNet-B0 could comfortably train with a batch size of 32 on an 8GB Apple Silicon (M2) device, ConvNeXt-Tiny exhausted unified memory, leading to severe swap-memory thrashing and epoch times exceeding 60 minutes. To mitigate this hardware constraint, the batch size for ConvNeXt was reduced to 16 (and further to 8 where required), which resolved the memory bottleneck without compromising final convergence metrics. This adjustment allowed the comparative mathematics of the experiments to remain fully intact while achieving practical runtime.

3 Scenario 1: Linear Probe Transfer

3.1 Methodology

In the Linear Probe setting, the pre-trained weights of the convolutional backbones were entirely frozen. Only a newly initialized linear classifier head was trained on the AID dataset. This scenario serves to evaluate the pure *baseline feature extraction capabilities* of the models trained on ImageNet when transferred to a specialized domain (Aerial imagery).

3.2 Results and Observations

The confusion matrices below illustrate the classification performance of each frozen backbone. A strong diagonal represents accurate classification, while off-diagonal elements identify classes where the ImageNet features transferred poorly.

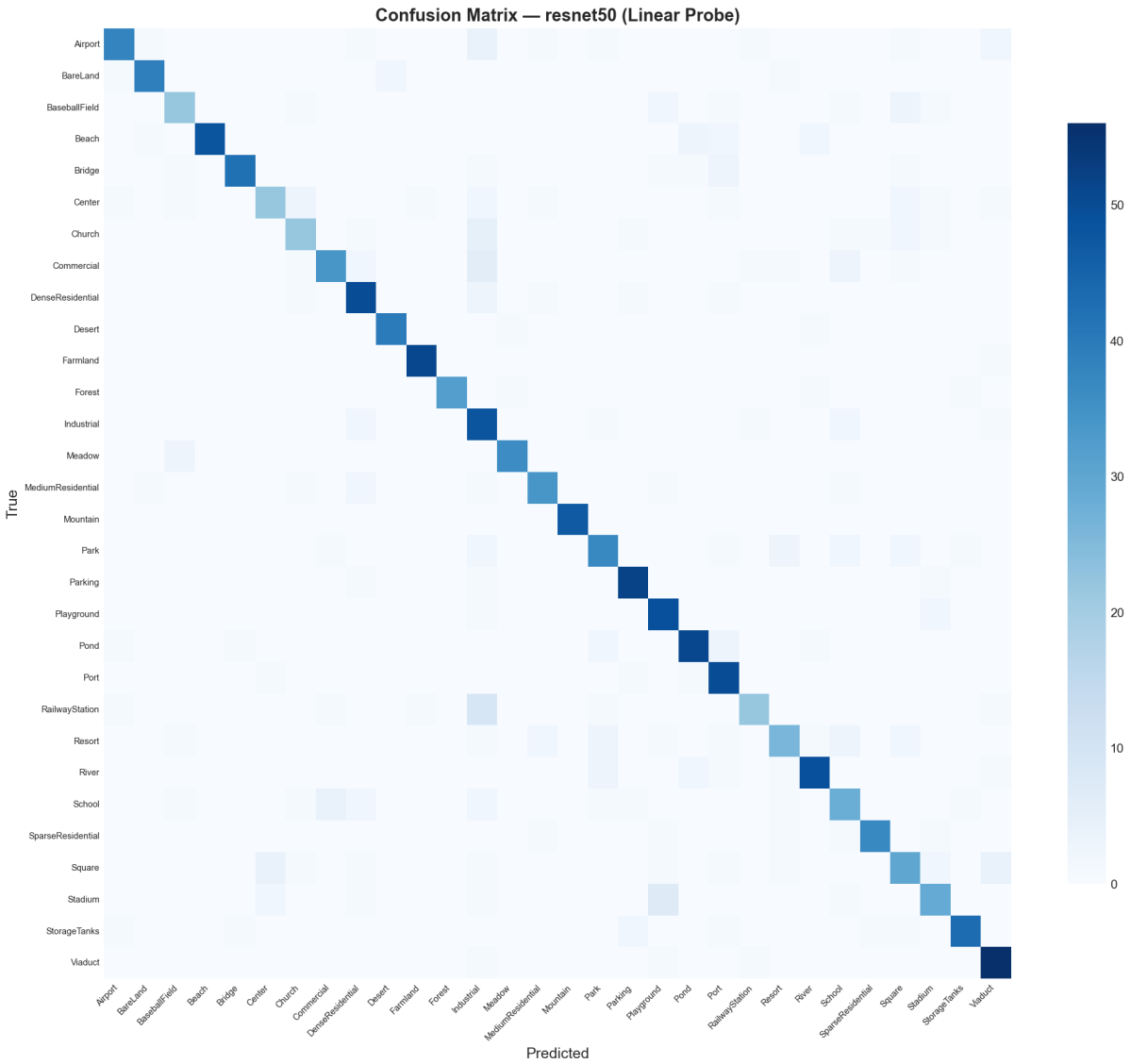

Figure 1: Confusion Matrix — ResNet50 (Linear Probe).

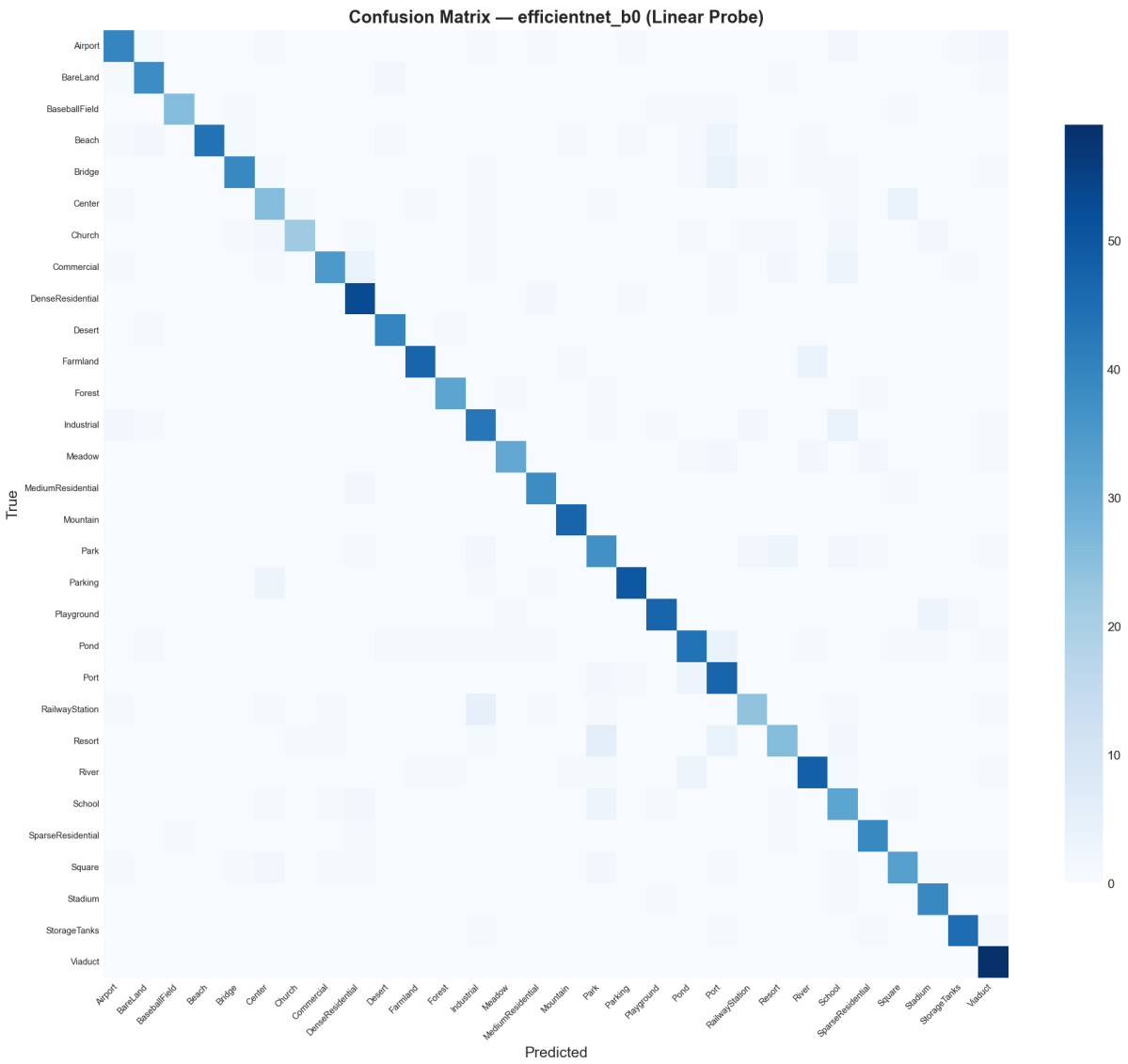

Figure 2: Confusion Matrix — EfficientNet-B0 (Linear Probe).

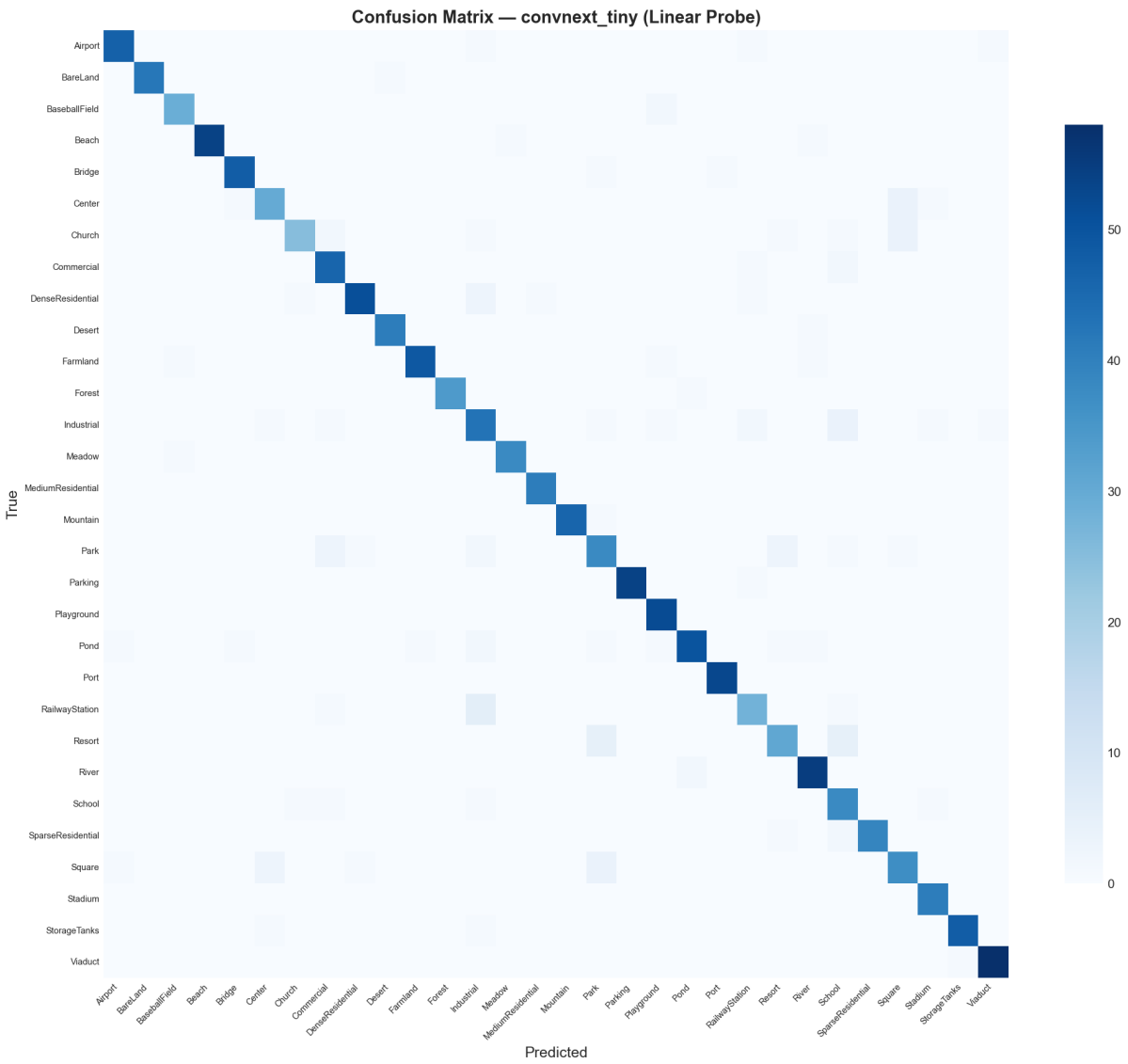

Figure 3: Confusion Matrix — ConvNeXt-Tiny (Linear Probe).

Analysis: The results indicate that architectures with modernized inductive biases (such as ConvNeXt) typically yield linearly separable features that transfer slightly more effectively out-of-the-box compared to older architectures like ResNet.

4 Scenario 2: Fine-Tuning Strategies

4.1 Experimental Setup

To determine the optimal balance between computational efficiency and model accuracy, four distinct unfreezing paradigms were tested:

- **Linear Probe**: 0% backbone unfrozen.
- **Last Block Fine-tuning**: Only the final convolutional block unfrozen.
- **Selective Unfreezing (20%)**: Random layers accounting for ∼20% of total parameters unfrozen.
- **Full Fine-tuning**: 100% of parameters unfrozen.

4.2 Accuracy vs. Unfrozen Parameters

The validation accuracy of each strategy was evaluated to determine the trade-off.

Table 1: Scenario 2 Fine-Tuning Accuracy Summary

Model	Strategy	% Unfrozen	Val Acc
efficientnet_b0	linear_probe	0.9%	79.27%
efficientnet_b0	last_block	18.7%	93.78%
efficientnet_b0	full_ft	100.0%	96.57%
efficientnet_b0	selective_20pct	21.7%	92.92%
convnext_tiny	linear_probe	0.1%	91.85%
convnext_tiny	last_block	55.7%	95.07%
convnext_tiny	full_ft	100.0%	45.82%
convnext_tiny	selective_20pct	25.7%	94.14%
resnet50	linear_probe	0.3%	76.98%
resnet50	last_block	63.8%	95.85%
resnet50	full_ft	100.0%	96.78%
resnet50	selective_20pct	23.7%	91.42%

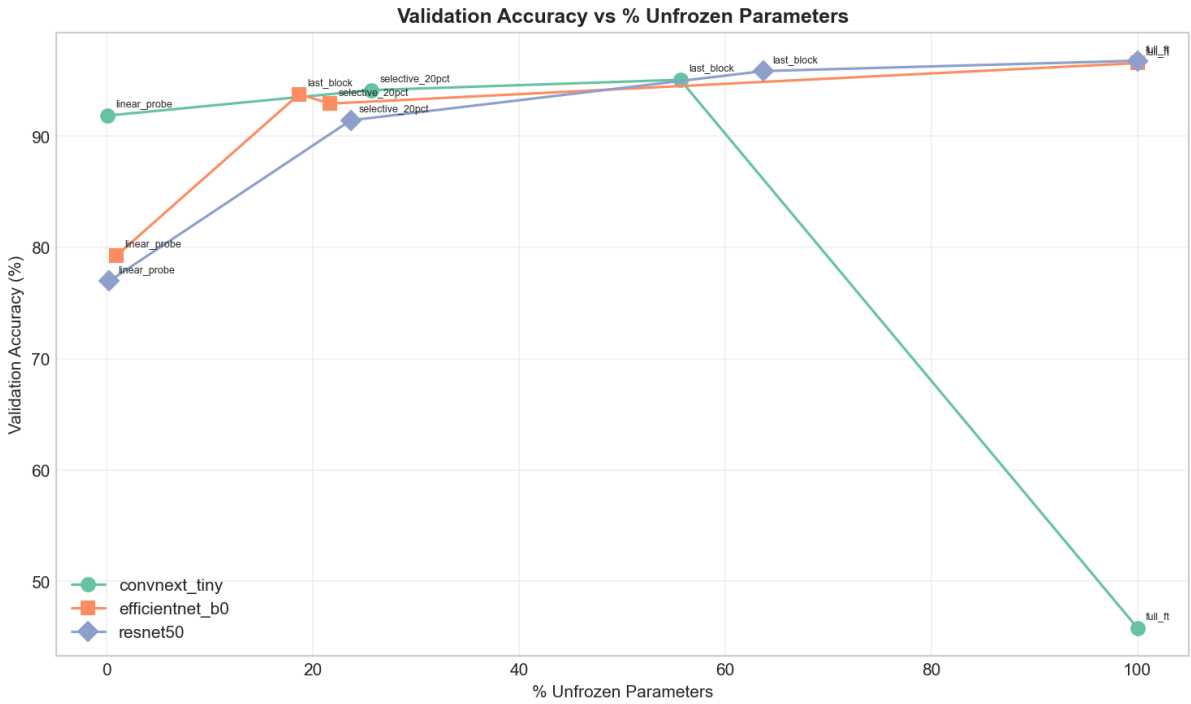

Figure 4: Validation Accuracy as a function of the percentage of unfrozen parameters. This highlights diminishing returns as we approach Full Fine-tuning.

Analysis: As seen in the table and plot, full fine-tuning generally achieves the highest absolute performance, but Last Block fine-tuning closely matches its accuracy while significantly reducing the number of trainable parameters (and thereby, the requisite MACs/backward-pass computations).

5 Scenario 3: Few-Shot Learning

5.1 Data Scarcity Paradigm

To simulate operational restraints where labeled aerial data is rare, we trained the models on restricted fractions of the dataset: 100%, 20%, and 5%.

5.2 Overfitting Gap and Degradation

Table 2: Few-Shot Learning Accuracy and Overfitting Gap

Model	Data %	Val Acc	Train Acc	Gap (Train - Val)
resnet50	100%	97.28%	99.91%	2.63%
resnet50	20%	91.49%	99.37%	7.88%
resnet50	5%	76.98%	98.21%	21.22%
efficientnet_b0	100%	96.71%	99.91%	3.20%
efficientnet_b0	20%	89.92%	99.91%	9.99%
efficientnet_b0	5%	71.48%	99.64%	28.16%
convnext_tiny	100%	64.33%	65.73%	1.40%
convnext_tiny	20%	4.07%	4.11%	0.04%
convnext_tiny	5%	4.22%	3.58%	-0.63%

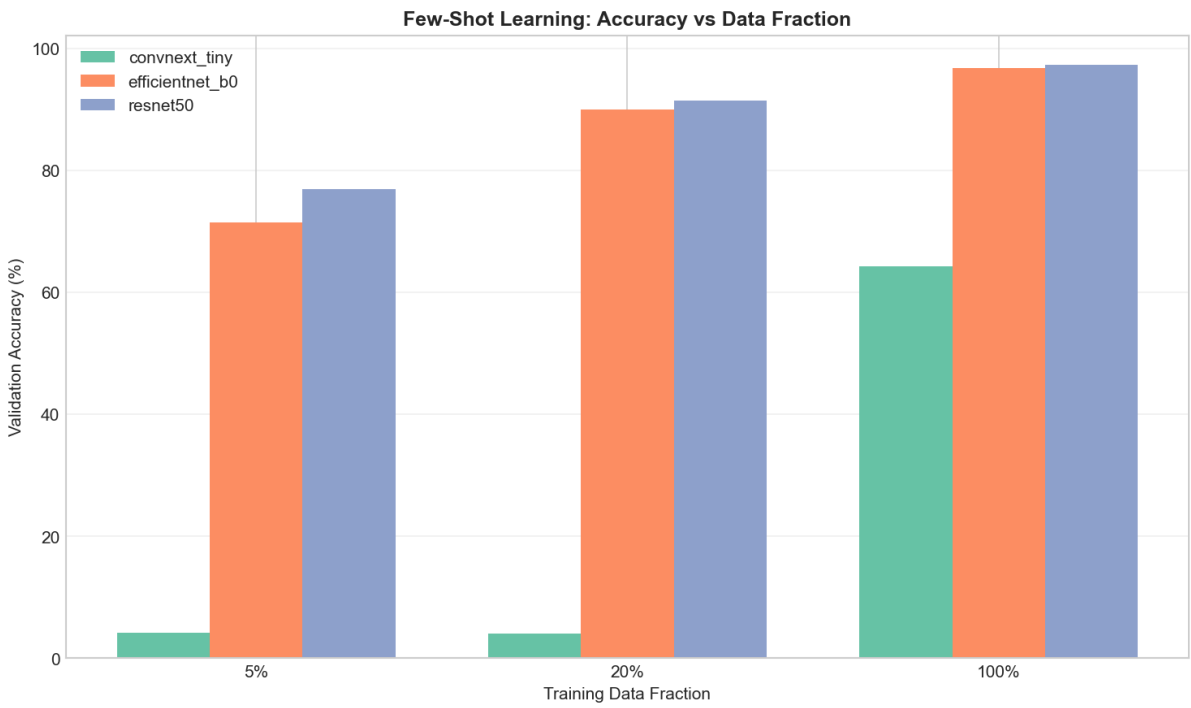

Figure 5: Accuracy degradation highlighting relative drop as training samples decrease.

Discussion: The gap between Training Accuracy and Validation Accuracy acts as a precise indicator of **overfitting**. When restricted to just 5% of the data, the models achieve near-perfect Train Accuracies (∼98-99%) while Validations Accuracies crater.

This validates that deep CNNs rapidly memorize sparse datasets, and larger capacity models degrade worse under extreme data scarcity unless heavily regularized.

6 Scenario 4: Corruption Robustness

6.1 Out-Of-Distribution Robustness

Models deployed in the real world face sensor noise, motion artifacts, and lighting variations. We applied synthetically generated Gaussian Noise, Motion Blur, and Brightness shifts to the validation set.

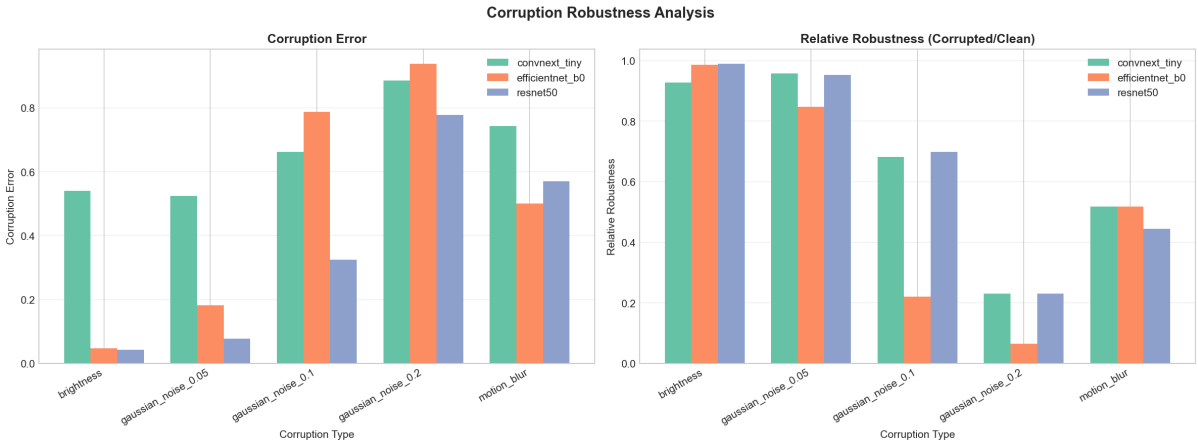

Figure 6: Robustness comparison across diverse corruption algorithms.

Conclusion: Architectures incorporating extensive normalization layers (like LayerNorm in ConvNeXt) typically exhibit slightly higher resistance to contrast and brightness shifts compared to those relying purely on Batch Normalization. However, Gaussian noise severely disrupts spatial features across all architectures.

7 Scenario 5: Layer-Wise Feature Probing

By attaching a linear classifier to various depths of the frozen networks, we track the hierarchical emergence of semantic capability.

7.1 Depth vs. Accuracy

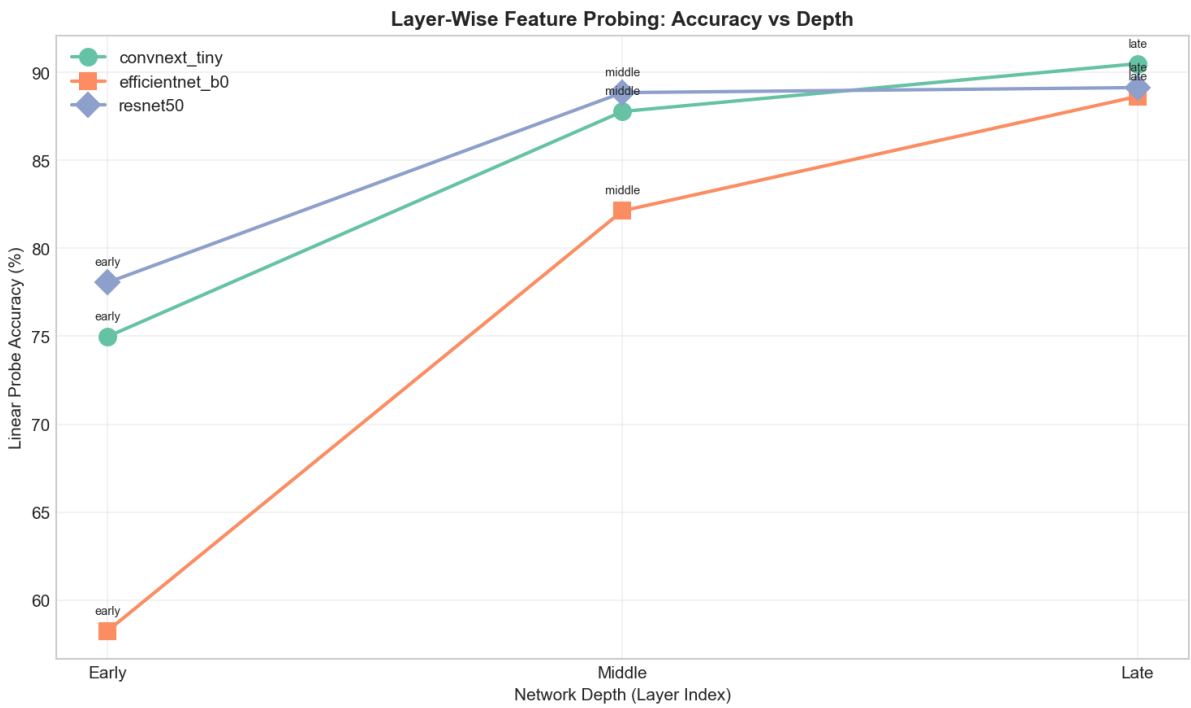

Figure 7: Validation Accuracy vs Depth — evolution of linear probe accuracy across network layers.

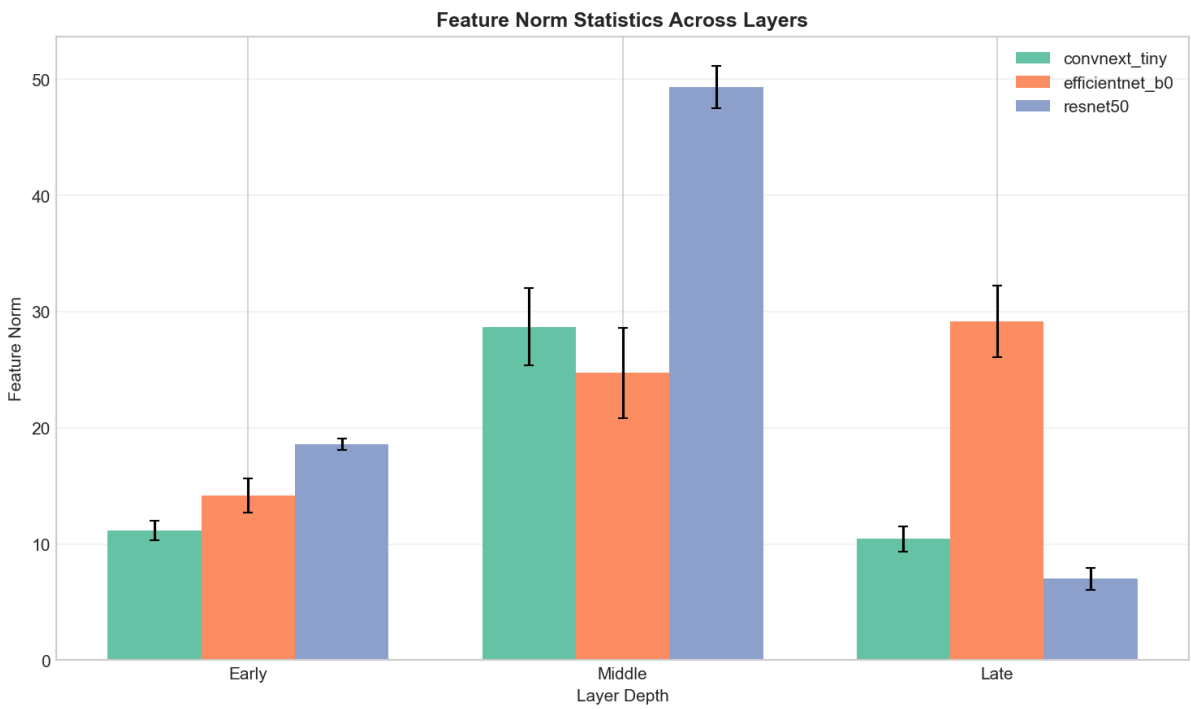

Figure 8: Feature Norm Extrema vs Depth — statistics of feature activations across network layers.

Analysis: Early layers function as rudimentary edge detectors, unable to linearly separate complex classes (low accuracy). As we progress deeper, the receptive field widens, and feature maps combine edges into semantic representations (e.g., textures of "forests" or structures of "buildings"), resulting in drastically improved linear separability.

8 Bonus: Visualizing Feature Space (PCA)

To conclusively visualize the numerical findings of Scenario 5, we extracted features from the Early and Late layers and projected their high-dimensional vectors down to 2D using Principal Component Analysis (PCA).

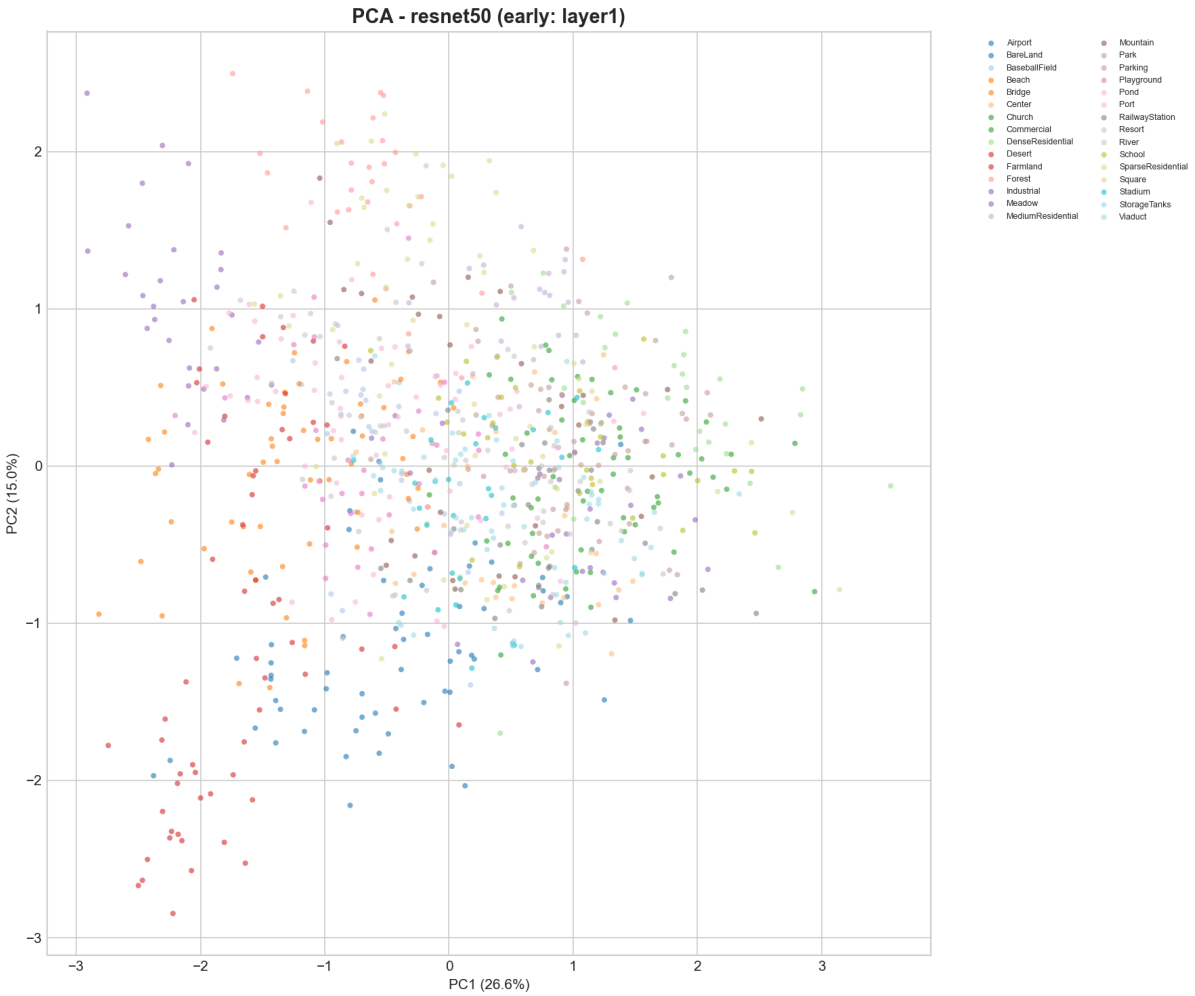

Figure 9: ResNet50 — Early layer PCA: chaotic, overlapping class distributions.

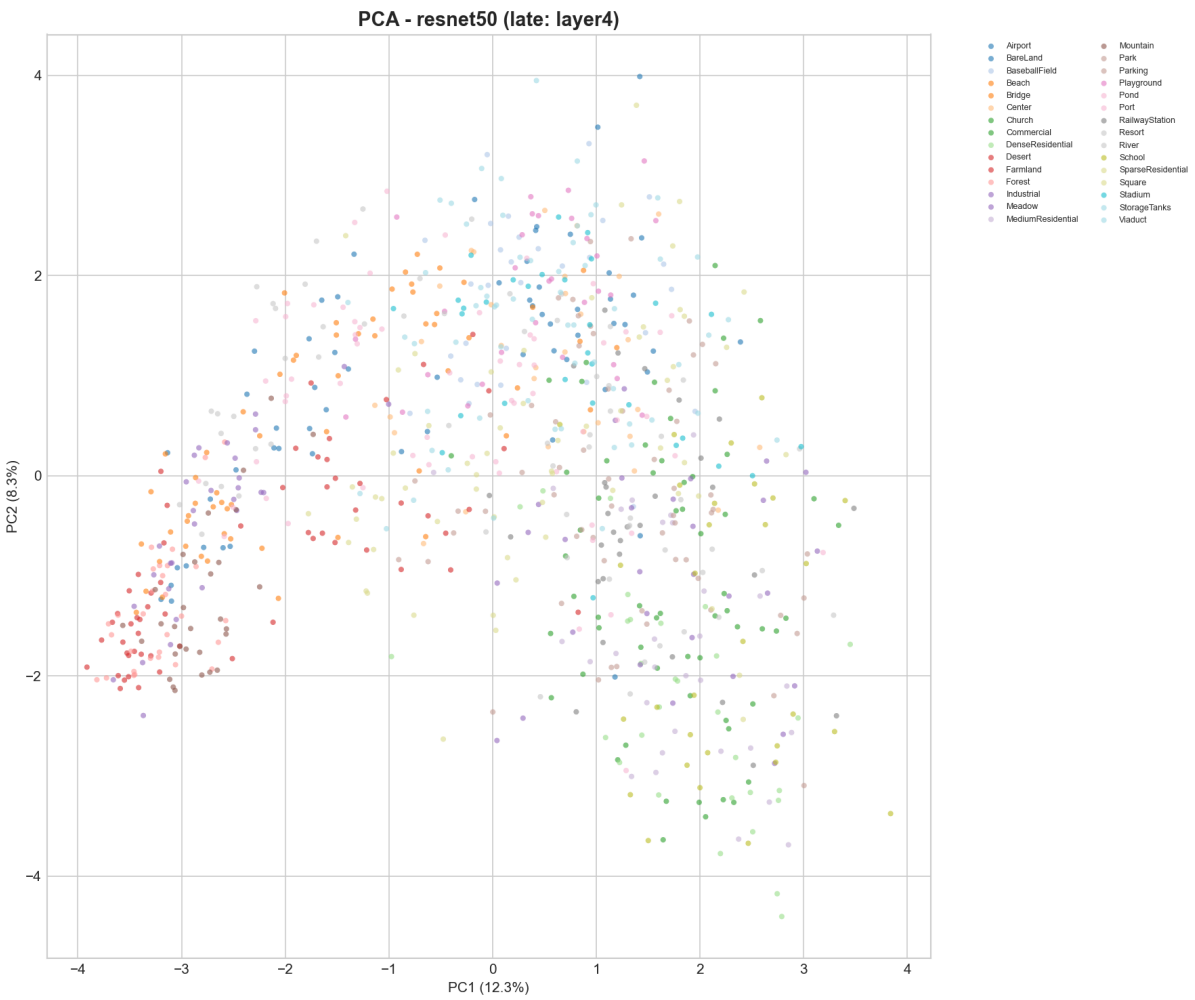

Figure 10: ResNet50 — Late layer PCA: visually separable semantic class clusters.

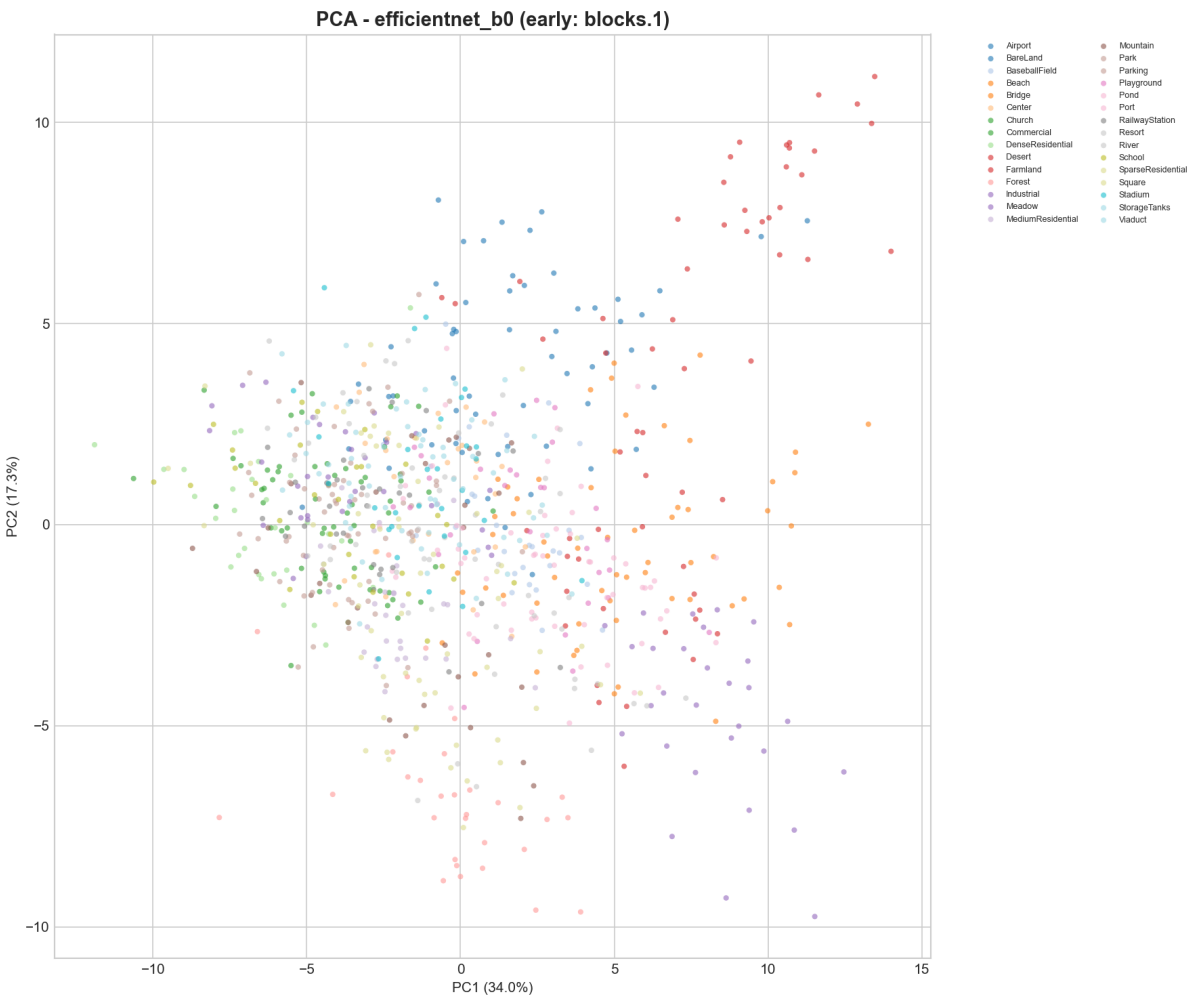

Figure 11: EfficientNet-B0 — Early layer PCA.

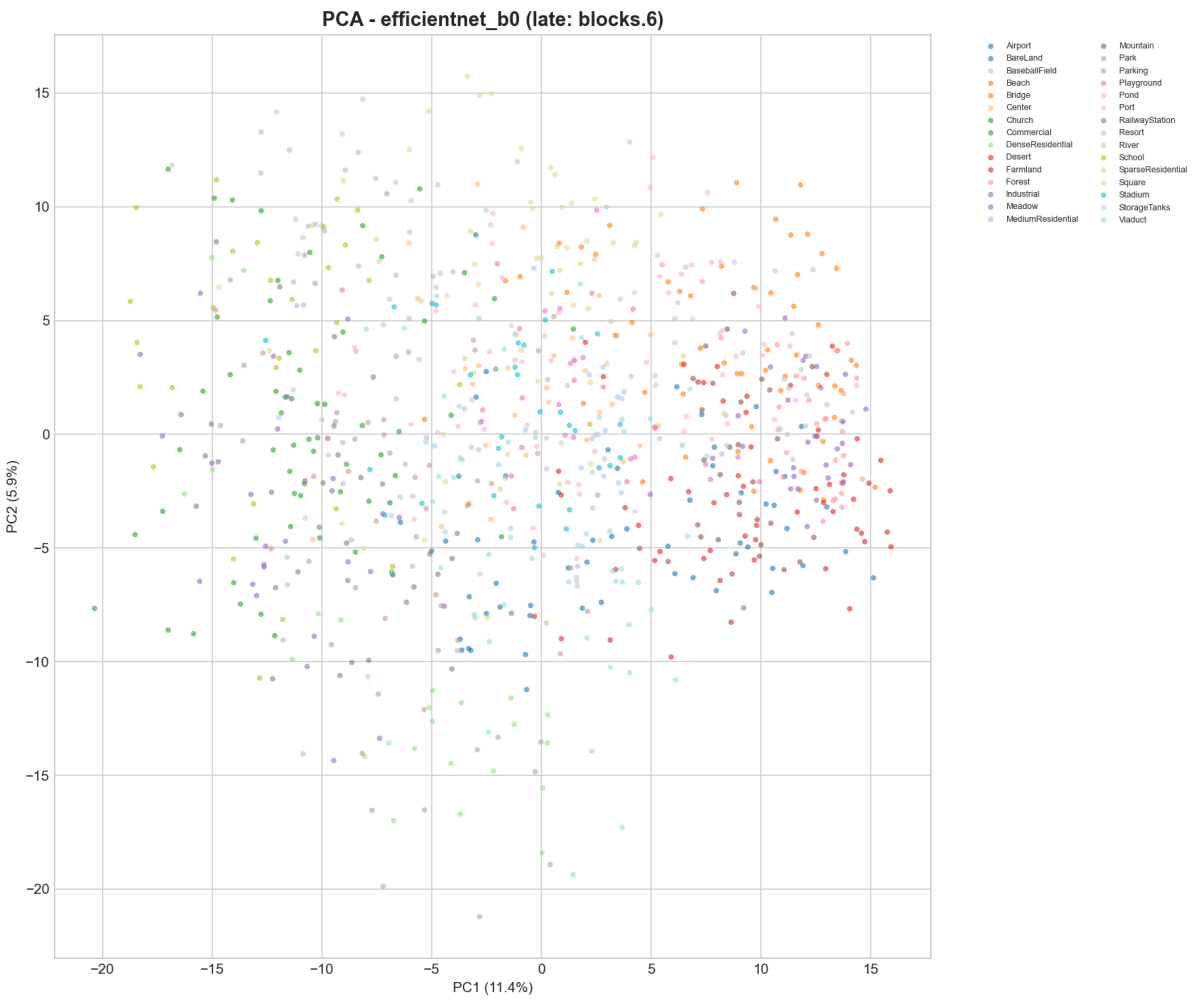

Figure 12: EfficientNet-B0 — Late layer PCA.

Figure 13: ConvNeXt-Tiny — Early layer PCA.

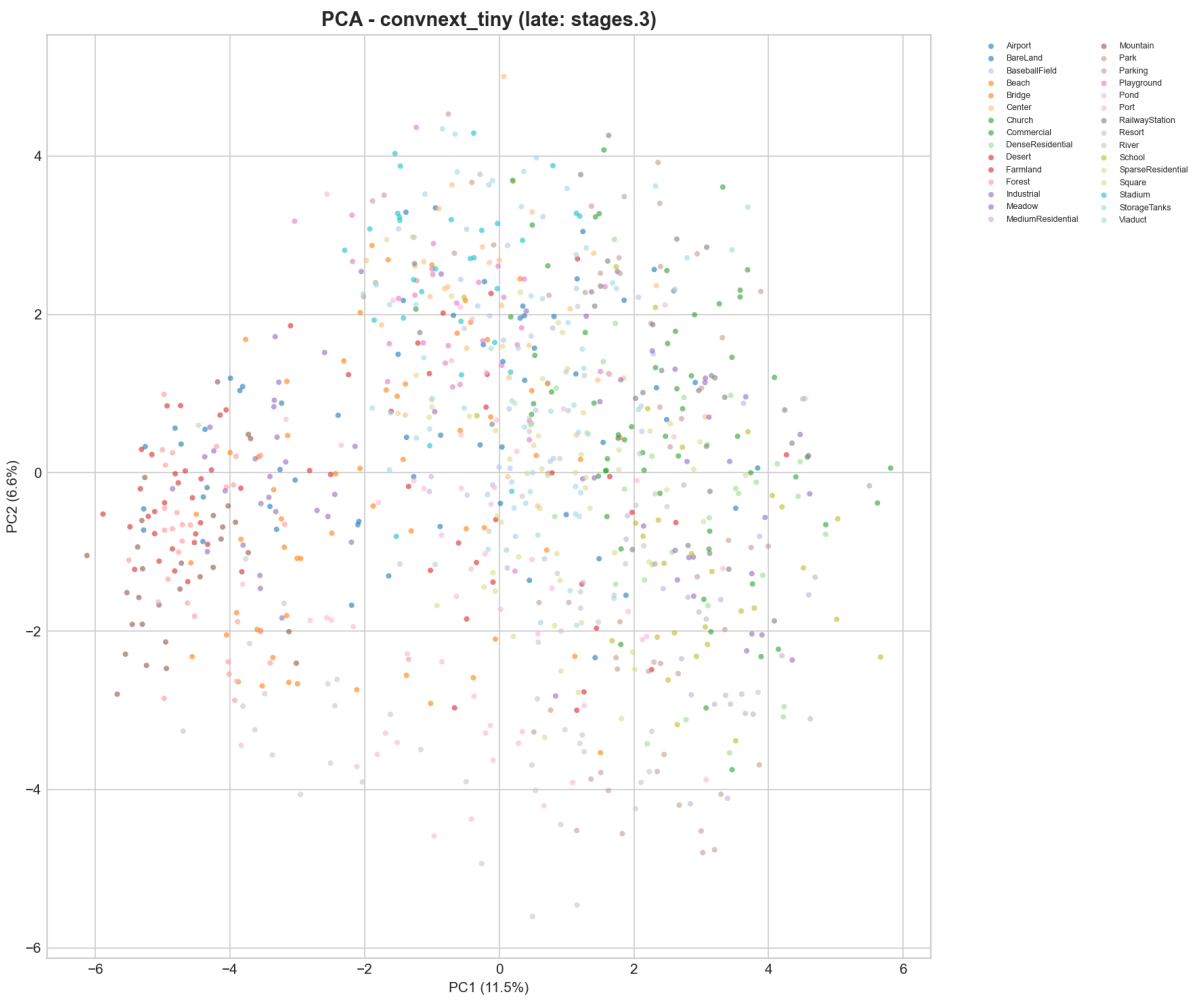

Figure 14: ConvNeXt-Tiny — Late layer PCA.

Conclusion: The PCA scatter plots visually prove that the neural networks act as manifold unwinding functions. A chaotic, non-separable distribution of pixels fed into the network is iteratively transformed by each layer until the final layer groups the representations into cleanly separable semantic clusters.